

PORVOO EMASALO, Finland

WMO: 029910

Lat: 60.204N

Lon: 25.626E

Elev: 89

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.8	3.8	-6.2	4.0	-1.5	-0.2	5.5	4.9	39.9	32.8	36.2	32.7	12.1	40	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	76.2	67.9	73.5	66.4	70.8	64.2	69.9	74.6	67.7	72.4	65.4	69.8	12.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	103.0	73.7	65.7	95.2	71.2	63.5	87.9	68.3	34.0	74.8	32.1	72.6	30.3	69.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.6	34.5	31.2	-4.0	79.7	7.7	5.0	-9.5	83.4	-14.0	86.3	-18.3	89.2	-23.9	92.8		
(5)			-4.5	69.9	7.6	2.6	-9.9	71.8	-14.4	73.4	-18.6	74.8	-24.2	76.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	25.0	24.0	29.2	38.4	49.5	57.2	64.3	62.9	55.2	44.9	37.5	32.2	(6)	(7)
	DBStd	15.36	9.96	9.95	7.02	4.79	6.61	4.90	5.37	4.47	4.92	5.92	6.79	7.82	(7)	(8)
	HDD50	3705	774	728	645	349	88	4	0	0	14	175	375	553	(8)	(9)
	HDD65	7949	1239	1148	1110	797	480	239	80	95	294	624	825	1018	(9)	(10)
	CDD50	1325	0	0	0	2	74	220	443	399	171	16	0	0	(10)	(11)
	CDD65	92	0	0	0	0	1	5	58	28	0	0	0	0	(11)	(12)
	CDH74	200	0	0	0	0	1	6	160	33	0	0	0	0	(12)	(13)
	CDH80	10	0	0	0	0	0	0	10	0	0	0	0	0	(13)	(14)
(14) Wind	WSAvg	15.7	17.2	15.1	15.6	14.1	13.7	14.3	13.1	14.3	15.7	17.0	18.8	20.0	(14)	(15)
(15) Precipitation	PrecAvg	26.5	2.2	1.7	1.4	1.4	1.5	2.6	2.5	2.9	2.4	2.8	2.9	2.3	(15)	(16)
	PrecMax	34.1	4.4	4.1	2.6	2.8	2.9	5.4	5.8	5.7	6.1	7.0	7.4	5.8	(16)	(17)
	PrecMin	21.3	0.3	0.1	0.3	0.3	0.2	0.7	0.0	0.2	0.5	0.8	0.4	0.6	(17)	(18)
	PrecStd	3.6	0.9	1.0	0.6	0.7	0.9	1.3	1.4	1.4	1.5	1.4	1.4	1.1	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	39.5	44.5	58.0	71.0	73.6	81.2	77.1	66.9	57.2	49.3	45.5	(19)	(20)
		MCWB	39.5	36.1	38.6	47.7	56.5	63.2	69.8	69.7	61.3	54.1	47.8	44.0	(20)	(21)
	2%	DB	39.3	36.5	40.9	51.2	66.5	69.7	77.9	74.3	64.7	55.5	48.1	43.7	(21)	(22)
		MCWB	38.4	35.0	36.7	43.8	57.1	61.2	68.7	67.5	60.5	53.8	47.1	42.2	(22)	(23)
	5%	DB	37.9	35.1	38.3	47.8	62.8	67.1	75.3	71.7	62.9	54.2	46.7	42.3	(23)	(24)
		MCWB	37.0	34.3	35.4	41.7	54.9	59.5	67.4	65.4	59.4	52.5	45.7	41.1	(24)	(25)
	10%	DB	36.3	34.1	36.5	45.0	59.4	64.6	72.9	69.5	61.6	52.7	45.4	40.7	(25)	(26)
		MCWB	35.5	33.5	34.6	40.4	52.9	57.8	66.4	64.0	58.6	51.0	44.3	39.6	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	36.8	39.5	48.4	61.1	66.3	72.5	71.4	63.2	55.6	48.5	44.6	(27)	(28)
		MCDB	40.5	38.6	42.7	54.8	66.3	70.4	77.9	76.1	65.9	56.1	48.7	45.4	(28)	(29)
	2%	WB	38.7	35.5	37.5	45.0	58.1	63.1	70.8	68.9	61.6	54.3	47.3	42.7	(29)	(30)
		MCDB	39.2	36.1	39.7	50.0	63.4	68.1	75.9	73.0	63.8	55.1	47.7	43.4	(30)	(31)
	5%	WB	37.2	34.6	36.2	42.8	56.1	60.7	69.2	66.7	60.4	52.9	45.9	41.2	(31)	(32)
		MCDB	37.8	35.2	37.8	46.4	61.7	65.3	74.1	70.7	62.2	53.8	46.5	42.1	(32)	(33)
	10%	WB	35.7	33.7	35.0	41.1	53.3	58.9	66.9	64.9	59.2	51.5	44.6	39.8	(33)	(34)
		MCDB	36.3	34.2	36.2	44.2	59.4	63.5	72.1	68.5	61.2	52.6	45.3	40.7	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.1	8.2	8.6	9.7	9.2	9.2	8.3	7.3	6.6	5.5	5.9	(35)	(36)
		MCDBR	5.1	4.7	8.3	12.6	13.9	12.6	12.0	10.3	8.4	6.1	4.8	5.3	(36)	(37)
	5% WB	MCWBR	5.6	4.3	5.9	7.5	8.0	7.1	6.6	6.2	5.8	5.4	5.2	5.4	(37)	(38)
		MCWBR	5.1	4.6	6.9	10.6	13.0	10.6	10.5	9.1	7.2	5.6	4.7	4.8	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.278	0.310	0.333	0.363	0.362	0.358	0.380	0.373	0.347	0.333	0.306	N/A	(39)	(40)
		taud	2.140	2.181	2.233	2.325	2.410	2.468	2.430	2.446	2.508	2.394	2.213	N/A	(40)	(41)
	Ebn at Noon		150	213	249	265	274	277	267	259	248	207	138	N/A	(41)	(42)
		Edh at Noon	14	23	30	33	33	31	32	30	23	19	13	N/A	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		78	260	733	1221	1667	1732	1679	1298	789	341	95	46	(43)	(44)
	RadStd		14	51	94	128	180	134	152	134	102	53	17	7	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	-266	-293	N/A	N/A

Nomenclature: See separate page